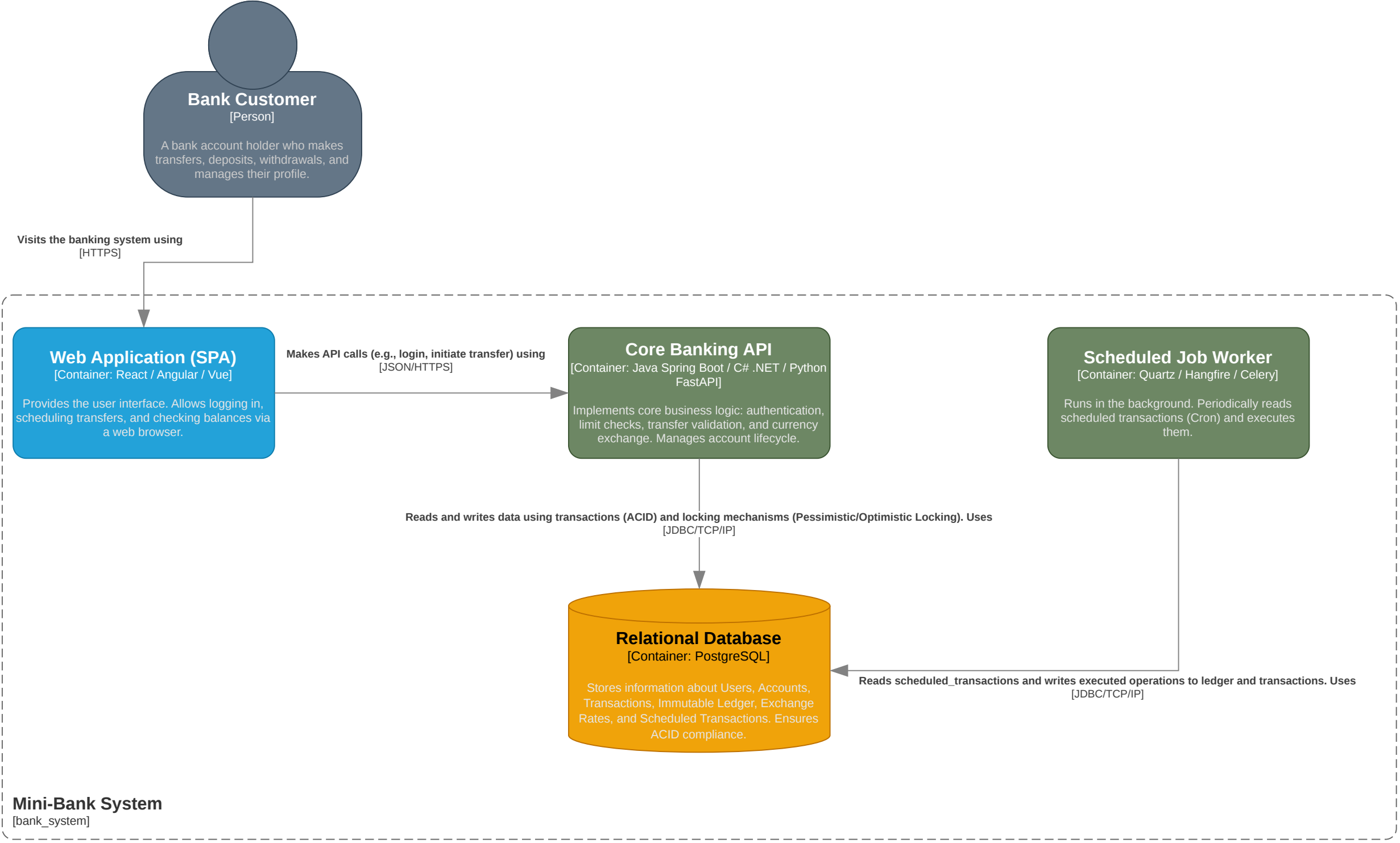
Bank Customer
[Person]
A bank account holder who makes transfers, deposits, withdrawals, and manages their profile.
Visits the banking system using
[HTTPS]
Web Application (SPA)
[Container: React / Angular / Vue]
Provides the user interface. Allows logging in, scheduling transfers, and checking balances via a web browser.
Makes API calls (e.g., login, initiate transfer) using
[JSON/HTTPS]
Core Banking API
[Container: Java Spring Boot / C# .NET / Python FastAPI]
Implements core business logic: authentication, limit checks, transfer validation, and currency exchange. Manages account lifecycle.
Scheduled Job Worker
[Container: Quartz / Hangfire / Celery]
Runs in the background. Periodically reads scheduled transactions (Cron) and executes them.
Reads and writes data using transactions (ACID) and locking mechanisms (Pessimistic/Optimistic Locking). Uses
[JDBC/TCP/IP]
Relational Database
[Container: PostgreSQL]
Stores information about Users, Accounts, Transactions, Immutable Ledger, Exchange Rates, and Scheduled Transactions. Ensures ACID compliance.
Reads scheduled_transactions and writes executed operations to ledger and transactions. Uses
[JDBC/TCP/IP]
Mini-Bank System
[bank_system]